

JAGIELLONIAN UNIVERSITY
IN KRAKOW

Lecture 1: Scientific Discovery: Understanding fr
From AI to Deep Learning

Deep Learning for Scientific Discovery

Marcin Płodzień

Atomic Optics Department
Institute of Theoretical Physics
Jagiellonian University in Kraków



Doctoral School
of Exact and
Natural Sciences
Jagiellonian University

1

Discovery from Data



JAGIELLONIAN UNIVERSITY
IN KRAKOW

The First Recorded Sky Observations



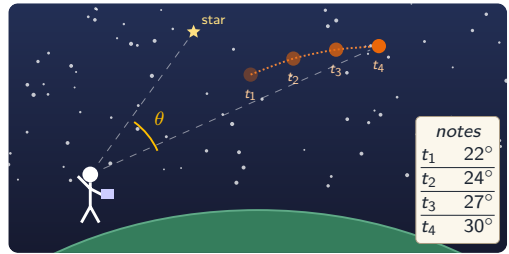
Venus tablet
c. 17th c. BCE (copy: 7th c. BCE)
21 years of Venus's first and last visibility at dawn and dusk



Shang oracle bones
China, 14th–12th c. BCE
records of solar and lunar eclipses



Astronomical diaries
Babylon, 652–61 BCE
about 600 years of nightly records



Someone on Earth watches the sky and writes down *how a position changes in time*.

Besides the sky: weather, river levels, market prices — **a 600-year dataset**. From it came **patterns**: an ≈18-year eclipse cycle used to *predict* eclipses — with no model of *why*.

Images: Wikimedia Commons — Venus tablet (British Museum), CC BY-SA 3.0; oracle bone (G. L. Todd), CC0; diary (British Museum), public domain.

Data, Laws and Theories in History



Babylon

c. 700–50 BCE

prediction

eclipses repeat every ≈ 18 years



Tycho Brahe

1576–1601

the data

angles on the sky at time t



J. Kepler

1609, 1619

the law: *how*

ellipses · equal areas

$T^2 \propto a^3$



I. Newton

1687

the theory: *why*

$\boldsymbol{F} = m\boldsymbol{a}$

$F = G\,Mm/r^2$



A. Einstein

1915

theory from ideas

$$G_{\mu\nu} = \frac{8\pi G}{c^4}\, T_{\mu\nu}$$

geometry of space-time = energy–momentum tensor

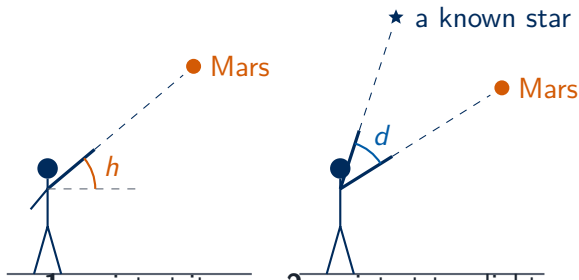
Our understanding of the **laws of nature** is *embedded in observations* — science is the art of extracting it.

What is new is who does the extracting: a *computer*, on data no human can see.

Images: Wikimedia Commons (public domain). Kepler: copy by A. Köhler (1910) of an early portrait. Einstein: F. Schmutzer (1921).

Sky Observations (1/2): Pointing at the Sky

Standing outside at night, you can only **point**:



Write down date, time, h, d — every clear night, for 20 years.
Tycho's instruments were giant, precise *protractors with sights* (mural quadrant for h, sextant for d).

Sky Observations (2/2): Angles on the Sky

Only directions — never how far away.

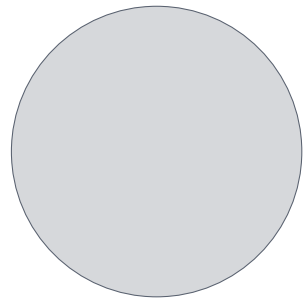
Angles on the sky

full circle $= 360^\circ$

$1^\circ = 60'$ arcminutes

$1' = 60''$ arcseconds

At arm's length: little finger $\approx 1^\circ$; the Moon $\approx \frac{1}{2}^\circ = 30'$.



Tycho's Instruments: Giant Protractors with Sights



Mural quadrant $\rightarrow h$, how high

A quarter-circle of brass (radius $\approx$2 m) on a wall facing exactly north–south. The observer (right) sights the star through an opening as it crosses due south. One assistant reads the *clock*, another *writes it down*: an early data pipeline.

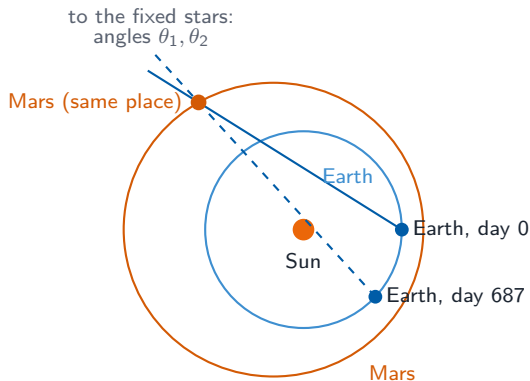


Sextant $\rightarrow d$, how far apart

Two sighting arms on a graduated arc: aim one at a known star, the other at the planet, and read the angle between them. This steel sextant was built so that a single observer could use it.

Both engravings: Tycho Brahe, *Astronomiæ instauratæ mechanica* (1598); Wikimedia Commons, public domain.

Kepler and Tycho's Angles (1/2): Triangulating Mars



sketch, not to scale; Mars eccentricity exaggerated

Tycho gave him *directions* only.

Kepler needed positions in space.

Kepler's trick (from 1600):

1. Mars returns to the *same point* every 687 days; Earth does not.
2. Two sight lines from two Earth positions $\Rightarrow$ **triangulate**: Mars's *position*, distance included.

Kepler and Tycho's Angles (2/2): Fitting the Orbit



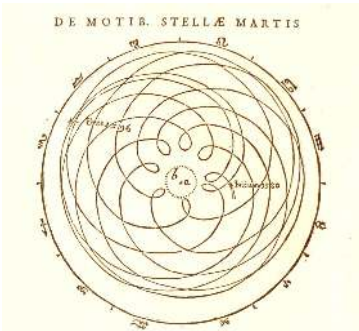
Tycho's mural quadrant, Uraniborg
(built 1582; engraving 1598)

Kepler's trick, continued:

3. The same trick in reverse first gives *Earth's* orbit.
4. Fit a curve to the reconstructed positions: circles miss Tycho by $8'$ $\rightarrow$ ovals $\rightarrow$ **ellipse**, Sun at a focus (c. 1605).

In today's words: *feature engineering* (angles $\rightarrow$ positions), then a model search.

Kepler's Model Search (1/2): Years of Hypotheses



Mars as seen from Earth, 1580–1596: Kepler's own drawing, *Astronomia Nova* (1609)

Raw data look like *this* — the simple law is hidden.

Years of hypotheses (1600–1605):

- circle + equant (an off-centre point from which the planet appears to move uniformly): fits Tycho's Mars *directions* to $2'$ — yet gives the wrong *distances*
- make it consistent with the distances $\Rightarrow$ the directions are off by **$8'$**, more than Tycho's error. Kepler trusts the *data*
- years of failed trials, egg-shaped ovals $\rightarrow$ finally the **ellipse** $\rightarrow$ three laws

Kepler's Model Search (2/2): Eight Arcminutes

> *"Now, because they could not be disregarded, these eight minutes alone will lead us along a path to the reform of the whole of Astronomy."*
>
> J. Kepler, *Astronomia Nova*, ch. 19 (1609)

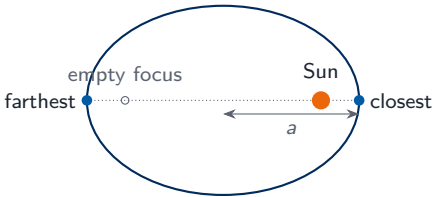
The discovery recipe — in 1600

data (Tycho's angles) → **representation** (triangulated positions) → **hypothesis class** (circles, ovals, ellipses) → **loss** (arcminutes of disagreement) → **law**.

Only the searcher has changed.

Kepler's Three Laws (1/2): Ellipses and Equal Areas

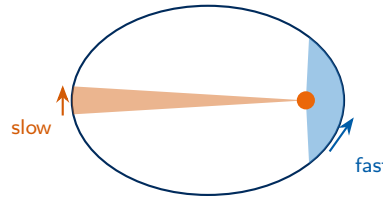
1. Law of ellipses (1609)



Each planet moves on an *ellipse*, with the Sun at one *focus* — not at the centre.

a = semi-major axis (the "average" distance). Real orbits are nearly circular; this one is exaggerated.

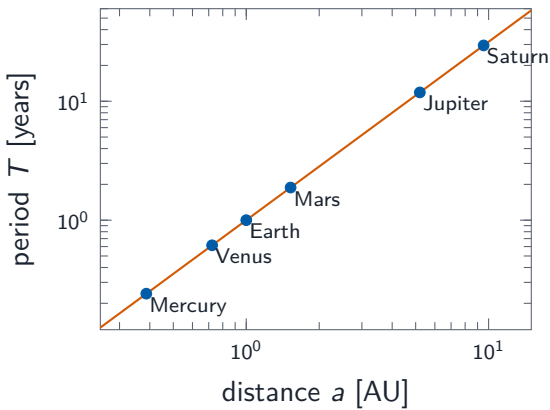
2. Equal areas (1609)



The line Sun–planet sweeps out *equal areas in equal times*.

Both shaded regions take the same time $\Rightarrow$ a planet moves **faster near the Sun**, slower far away.

Kepler's Three Laws (2/2): the Harmonic Law

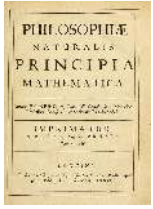


3. Harmonic law (1619)

$T^2 \propto a^3$: the farther the planet, the longer its year — by one rule for *all* planets.

In years and AU: $T^2 = a^3$. The six planets known to Kepler fall on one straight line (modern values).

Newton's Theory (1/2): Gravitation



Principia (1687), public domain

1684: Halley asks what orbit an inverse-square pull gives. Newton replied at once: an ellipse — he had calculated it (de Moivre's account).

The laws of motion alone are not enough: they say how bodies move *under* a force, not *which* force acts. Newton needed:

- laws of motion, $\boldsymbol{F} = m\boldsymbol{a}$
- **universal gravitation**, $F = G\,Mm/r^2$
- calculus, to solve for the orbit

Kepler's laws become consequences:

2nd $\leftarrow$ angular momentum conservation (any central force)

1st $\leftarrow$ specifically the $1/r^2$ force

3rd $\leftarrow$ $T^2 = \frac{4\pi^2}{G(M+m)}\,a^3$ — even *corrected*

Newton's Theory (2/2): Beyond the Data

The theory reaches beyond the data:

- *unifies* the apple, the Moon, the tides, the planets
- explains *deviations* from Kepler (Jupiter–Saturn pull)
- *predicts the unseen*: Halley's comet returns (1758); Neptune found from Uranus's anomalies (Le Verrier, 1846)

data (Tycho) → laws (Kepler) → theory (Newton): each level compresses the one below.

Kepler found **how** planets move; Newton's laws of motion *plus gravitation* explained **why** — and **predicted what no one had yet observed**.



U. Le Verrier; Wellcome Collection, CC BY 4.0

But Mercury's orbit still did not fit → Einstein.
Residuals are where discoveries live.
(residuals: small mismatches between data and model)

Einstein: Gravity from Principles and Mathematics (1/2)

Starting assumptions — not data:

1. **Equivalence principle** (1907): locally, gravity cannot be told apart from acceleration — inside a closed, freely falling lift you feel no weight.
2. **General covariance**: the laws of physics take the same form in *every* coordinate system — the same laws for every observer, however they move or label space and time.
3. **Newton as a limit**: for weak gravity and slow motion the theory must give back $F = G\,Mm/r^2$.

The mathematics was waiting:

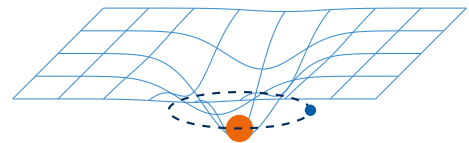
Riemann's geometry of curved spaces (1854), Ricci & Levi-Civita's tensor calculus (1900) — brought to Einstein by his friend M. Grossmann (1912).

Einstein: Gravity from Principles and Mathematics (2/2)

Einstein, 25 November 1915:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

curvature of space-time | energy & momentum of matter



cartoon: mass curves space-time; planets follow the curvature

"Spacetime tells matter how to move; matter tells spacetime how to curve."
— J. A. Wheeler, *Geons, Black Holes and Quantum Foam*

No data fitted. The only constant, G, is fixed by the Newtonian limit — yet Mercury's 43″ comes out. *The beauty of mathematics.*

Two Roads to a Law (1/2): Data and Principles

Road 1 — bottom-up: data → law

Tycho's angles → Kepler's ellipses. Patterns found *in* the data; Newton then explains them with a force law.

Machine learning lives here: it is superb at finding what is already hidden in the data.

Road 2 — top-down: principle → prediction → data

Einstein (1915) did *not* fit any data. He started from a concept — gravity is the *geometry of space-time* (equivalence principle) — and derived the consequences. It explains Mercury with nothing tuned — and its *new* predictions were confirmed only afterwards.

Science needs both roads: a tiny residual ($8'$, $43''$) breaks a theory — the next one may have to come from *ideas*, not fitting.

Two Roads to a Law (2/2): Mercury and New Predictions

Mercury: where Road 1 ran out

- 1859: Le Verrier finds Mercury's orbit turns (precesses: the ellipse slowly rotates, so the point closest to the Sun, the *perihelion*, shifts) $\sim 38''$ *per century* faster than Newton allows (refined to $43''$ by Newcomb, 1882).
- The Neptune trick — "add an unseen planet" — fails: the planet *Vulcan* is never found.
- General relativity gives exactly $43''$, *with no parameter tuned* to Mercury.

Then genuine predictions, later confirmed:

- starlight bent by the Sun — eclipse of 1919
- clocks run slower in gravity — GPS corrects for it daily
- gravitational waves — predicted 1916, detected 2015 (LIGO)

Falsifiability (1/2): Popper's Criterion

Falsifiability — K. Popper, *Logik der Forschung* (1934)

A model is scientific only if its predictions are *compared with observations* — and *could fail*. No number of agreements proves it; one solid disagreement refutes it.

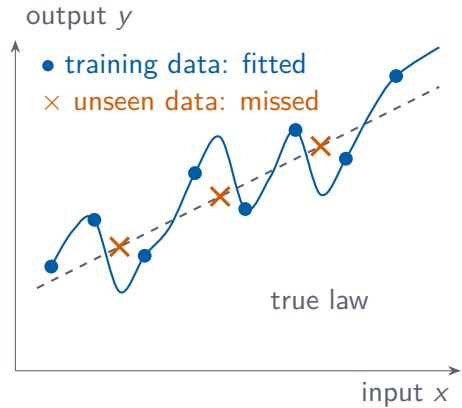
Our story is a chain of falsifications:

- circles ← refuted by $8'$ (Kepler)
- Newton ← refuted by $43''$ (Mercury)
- general relativity *survived* a risky test: the 1919 eclipse — Popper's own favourite example

Understanding = *predictive power*: **predict first**, *then* observe, then check.
Neptune (predicted 1846, then seen) · light bending (predicted 1915, seen 1919) · gravitational waves (predicted 1916, seen 2015)

Falsifiability (2/2): the Same Rule for Machine Learning

- fitting the training data is *not* evidence — Kepler's circle fitted Tycho to $2'$ and was still false
- the test: predictions on **data the model has never seen**
- a model that cannot be checked against observations is not yet science

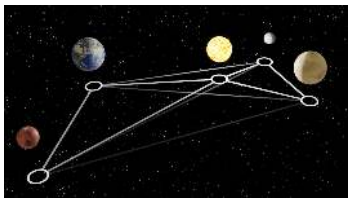


Machine Learning in This Story

	who searches	data	what comes out
Babylon	human	sky records, by eye	**prediction** — no mechanism
Kepler	human	Tycho's tables	**a compact law** — *how*
Newton / Einstein	human	laws, principles	**a theory** — *why*
Today	**machine**	10^6–10^{12} numbers, beyond human perception	usually **prediction** — a "Babylonian" black box

- The scientist does not disappear — they **design the search**: data, model class, loss, validation.
- Getting from *Babylon* to *Kepler* — reading a law out of a trained network — is the job of **interpretability** and **symbolic regression** (computers searching for a formula).

A Machine Rediscovers Newton's Gravity (1/2): the Pipeline

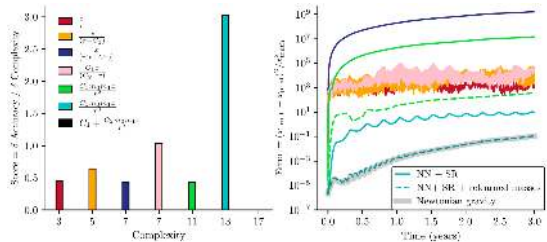


bodies = graph nodes; edges = learned interactions

1. **Data:** 30 years of trajectories of the Sun, planets and large moons.
2. A **graph neural network** learns to simulate them (given: Newton's 2nd & 3rd laws, symmetries).
3. **Symbolic regression** — a computer search over formulas — reads the learned force out as an equation.

P. Lemos *et al.*, Mach. Learn.: Sci. Technol. **4**, 045002 (2023), arXiv:2202.02306. Figure CC BY 4.0.

A Machine Rediscovers Newton's Gravity (2/2): the Formula



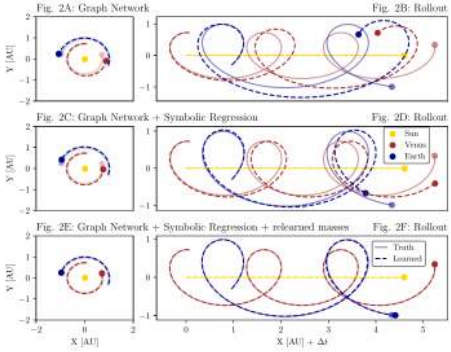
Left: candidate formulas, scored by accuracy gain per unit of complexity. Right: orbit error of each formula over 3 years.

Winner: $F_x \propto m_0 m_1 \, x / r^3$, i.e. $\boldsymbol{F} \propto m_0 m_1 \, \hat{\boldsymbol{r}} / r^2$ — **Newton's gravity**. The *masses* of planets and moons were inferred too.

The laws of motion were given — the machine found the missing piece: *exactly Newton's step*.

P. Lemos, N. Jeffrey, M. Cranmer, S. Ho, P. Battaglia, *Rediscovering orbital mechanics with machine learning*, Mach. Learn.: Sci. Technol. **4**, 045002 (2023), doi:10.1088/2632-2153/acfa63; arXiv:2202.02306. Figures CC BY 4.0.

Validation of the Discovered Law



Sun, Venus, Earth: truth (solid) vs. learned (dashed). Top: network alone. Middle: the found formula. Bottom: formula + masses re-estimated.

Figure from Lemos et al., Mach. Learn.: Sci. Technol. **4**, 045002 (2023), arXiv:2202.02306, CC BY 4.0.

- The *network alone* drifts off the true orbits.
- The **extracted formula** with re-estimated masses tracks them almost perfectly — *the law generalises better than the network it came from.*
- Masses of tiny moons stay poorly determined: they barely pull on anything, so the data *cannot* constrain them (unidentifiable parameters)

Kepler's 8 arcminutes, 400 years later: validation decides.

2

New Resources: Classical Data & Quantum Correlations



JAGIELLONIAN UNIVERSITY
IN KRAKOW

Every Revolution Starts with a New Resource

Age of Energy



Mechanical — wind, water, muscle

Each new resource, once harnessed, *reshaped society*.

Every Revolution Starts with a New Resource

Age of Energy



Heat — coal & the steam engine

Each new resource, once harnessed, *reshaped society*.

Every Revolution Starts with a New Resource

Age of Energy



Oil — dense, portable fuel → cars & flight

Each new resource, once harnessed, *reshaped society*.

Every Revolution Starts with a New Resource

Age of Energy

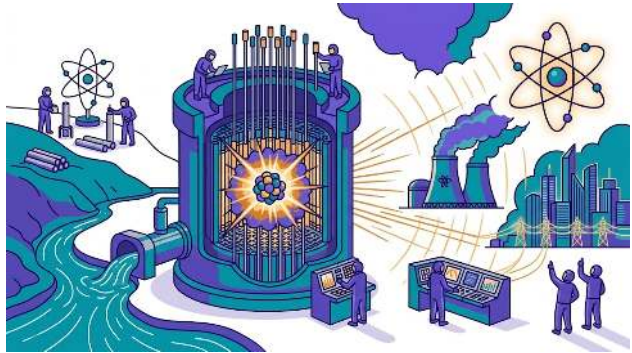


Electricity — power & the grid

Each new resource, once harnessed, *reshaped society*.

Every Revolution Starts with a New Resource

Age of Energy



Nuclear — energy from the atomic nucleus

Each new resource, once harnessed, *reshaped society*.

Every Revolution Starts with a New Resource

Age of (Classical) Information



Radio & the EM spectrum — wireless communication

Shannon (1948) made information a *measurable* resource — the *bit*.

Every Revolution Starts with a New Resource

Age of (Classical) Information



Electronics & logic — the computer

Shannon (1948) made information a *measurable* resource — the *bit*.

Every Revolution Starts with a New Resource

Age of (Classical) Information



Data — the fuel of machine learning

Telescopes, sequencers, satellites, simulations: science now produces more data than humans can read.

Every Revolution Starts with a New Resource

Age of (Quantum) Information



Quantum states — coherence & entanglement

From Shannon's *bit* to the **qubit** — quantum correlations as a *resource*.

Quantum Technologies: Four Pillars, One Foundation



One foundation: the **non-classical resources** — superposition, entanglement, coherence.

Four fields that harness them:

- **Computing** — classically intractable problems (factoring, optimisation, chemistry)
- **Simulation** — quantum matter & molecules
- **Communication** — secure keys (QKD), quantum internet
- **Metrology & sensing** — ultra-precise clocks & sensors

The 21st Century: Classical Data + Quantum Correlations



Data — the new raw material

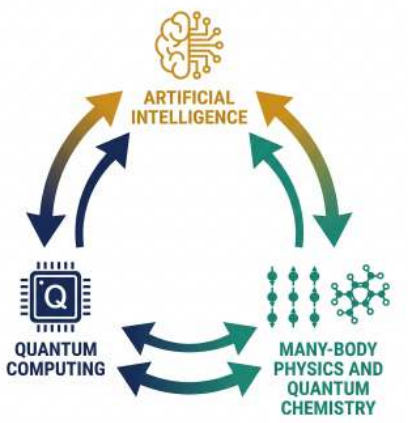
Images, sequences, sensor streams, simulations.
Machine learning extracts the patterns.

Quantum correlations — nature's deepest rules

Entanglement, coherence, exponential Hilbert space.
Quantum computers harness them.

The two engines **intersect**: ML helps build and control quantum devices; quantum devices may become learning machines.

Three Converging Fields



AI, **quantum computing** and **many-body physics & quantum chemistry** — deeply interconnected:

- AI $\rightarrow$ designs, controls and reads out quantum devices
- quantum computers $\rightarrow$ simulate molecules & materials beyond classical reach
- both $\rightarrow$ a shared target: **discovery of new materials, molecules and drugs**

The hope: the two technologies **merge** into one engine of scientific discovery.

Science Is Drowning in Data

- **Biology** — genomes, single-cell atlases, >200 M predicted protein structures
- **Chemistry & materials** — millions of simulated molecules and crystals
- **Earth sciences** — decades of satellite imagery & weather reanalysis (ERA5)
- **Astronomy** — sky surveys producing terabytes per night (Vera Rubin Observatory)
- **Physics** — particle collisions, simulations of turbulence and quantum matter
- **Mathematics** — vast search spaces of algorithms, constructions, proofs

The bottleneck has moved

from *collecting* data
$\rightarrow$ to *understanding* it.

Deep learning is the most successful tool we have for extracting structure from **high-dimensional** data — where intuition and classical statistics stop working.

Where is the bottleneck in your field?

2024: Two Nobel Prizes for Artificial Intelligence

Nobel Prize in Physics 2024

John J. Hopfield & **Geoffrey E. Hinton**

"for foundational discoveries and inventions that enable machine learning with artificial neural networks"

Hopfield network (1982) — memory as an energy landscape; Boltzmann machine (1985) — statistical physics of learning.

Nobel Prize in Chemistry 2024

David Baker — "for computational protein design"

Demis Hassabis & **John Jumper** — "for protein structure prediction"

AlphaFold: a 50-year-old problem of biology solved by a deep neural network.

Physics gave ideas *to* AI · AI gave answers *to* chemistry and biology.

The traffic goes both ways — and across all disciplines.

3

Artificial Intelligence, Machine Learning, Deep Learning



JAGIELLONIAN UNIVERSITY
IN KRAKOW

First, Some Big Open Questions

Before we build anything, it is worth pausing on questions we *cannot* fully answer:

- What is **(human) intelligence**?
- Is intelligence **computable**?
  i.e. could it, *in principle*, be reduced to an *algorithm* — a fixed set of explicit rules a machine mechanically follows (the sense A. Turing made precise)?
- Is intelligence the same as **consciousness** — the *subjective*, first-person experience of being aware — or just information processing?
- What do we even mean by **"Artificial Intelligence"**?
- Can machines that are *not* hard-coded **improvise, adapt, and solve** complex tasks — like humans?

Mathematicians, philosophers of science, biologists: you have a say here.

Artificial Intelligence

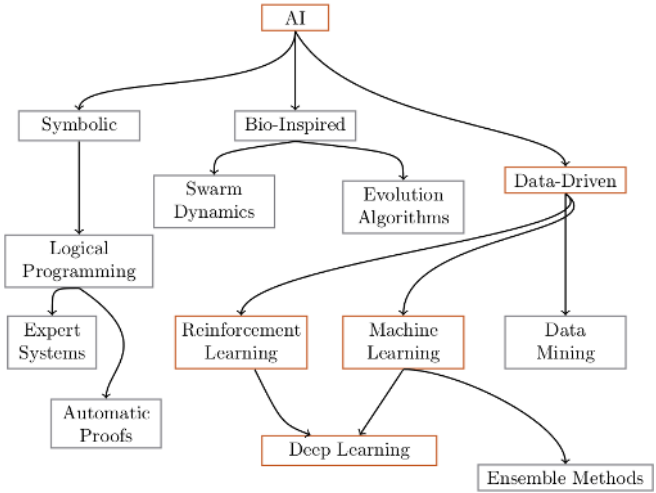
AI: systems that exhibit *intelligent behaviour*.

Biology-inspired approaches:

- Evolutionary algorithms — natural selection
- Swarm intelligence — ants & bees ▷ particle swarm optimisation · ants carrying a large object
- Cellular automata — emergent behaviour ▷ Conway's Game of Life

Modern dominant paradigm:
Data-driven learning — machine learning & *neural networks* discover the rules directly from data.
(this is where deep learning lives — our focus for the rest of the course)

The AI Landscape (1/2)



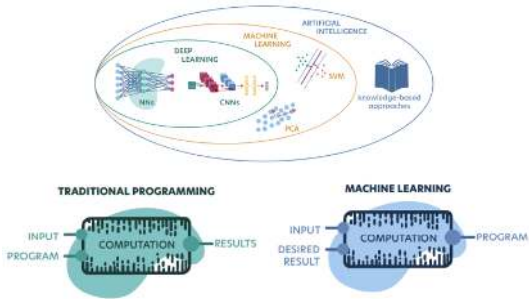
The AI Landscape (2/2): Three Paradigms

Three AI paradigms:

- **Symbolic** — rules and logic
  - logic programming (Prolog), expert systems
  - computer algebra
  - **proof assistants** (Lean): every step of a proof checked by machine (later today)
- **Bio-inspired** — evolution, swarms
  ▷ particle swarm optimisation · ants carrying a large object
- **Data-driven** — ML, DL, RL
  the dominant paradigm today — this course

Modern breakthroughs often *combine* them: AlphaProof = neural network + formal proof checker.

AI Hierarchy Nowadays



Each level is a *subset* of the one above:

- **AI** — any system with intelligent behaviour
- **ML** — learns its rules from *data*
- **DL** — ML with *deep neural networks*

"Generative AI", "foundation models", "LLMs" all live inside **DL**.

Traditional vs Data-Driven Programming













German Shepherd Dachshund



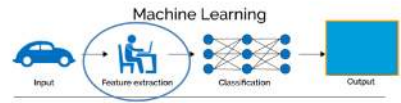
Algorithm-driven

Quite difficult to hard-code algorithm for image classification

Traditional: two breeds — hard-coding an image classifier is very hard.

Traditional vs Data-Driven Programming



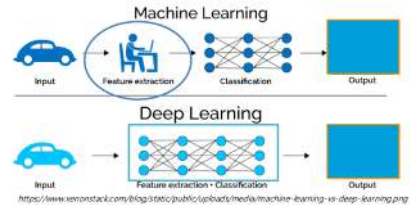


- length/height; ratio lenght/height;
- ears position; color; etc

Machine Learning: learn from examples — but *you* still hand-pick the features (length, ears, colour. . .).

Traditional vs Data-Driven Programming





Crucial features are extracted automatically

Deep Learning: the network extracts the crucial features *automatically*.

The Same Shift — in Science

Theory-driven

rules → predictions

Newton's laws, Schrödinger equation, reaction kinetics, population models.

Kepler's laws from Newton.

Classical data analysis

data + chosen model → parameters

Fitting a line, a Gaussian, an exponential decay — *you* choose the formula.

Fitting G, or a decay rate, once the law is known.

Deep learning

data → the model itself

A flexible function with millions of parameters; the *form* is learnt too.

No formula needed — but also no formula given. Kepler *searched* for the form himself.

Every scientist already does **machine learning** when fitting a curve.
Deep learning is curve fitting with **enormously flexible curves** — the art is keeping them from fitting the noise.

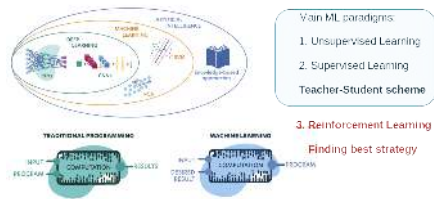
Three Main ML Paradigms (1/2)

1. Supervised Learning

- labelled data $\rightarrow$ learn a mapping $f : X \rightarrow Y$
- classification, regression
  e.g. microscopy image $\rightarrow$ cell type; molecule $\rightarrow$ energy

2. Unsupervised Learning

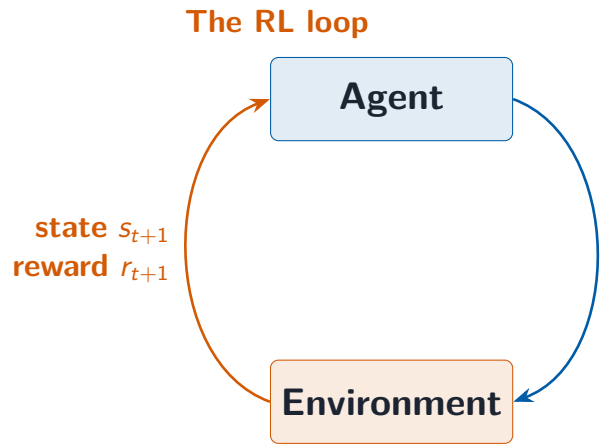
- unlabelled data $\rightarrow$ discover structure
- clustering, dimensionality reduction, generation
  e.g. new cell types in single-cell data; phases of matter



Three Main ML Paradigms (2/2)

3. Reinforcement Learning

- agent $\leftrightarrow$ environment $\rightarrow$ maximise reward
- games, robotics, *control of experiments*
  e.g. plasma control in a tokamak; algorithm discovery



4

The Recipe of Deep Learning

JAGIELLONIAN UNIVERSITY
IN KRAKOW

Neural Network Anatomy

⚙ under the hood

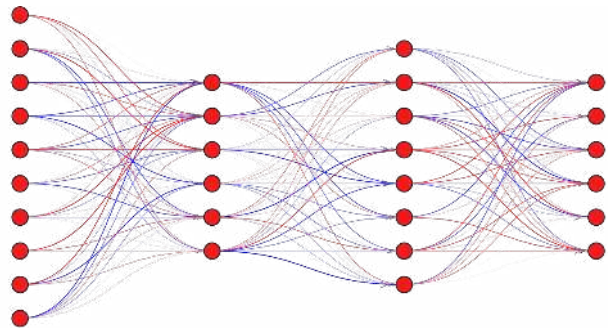
Non-linear, parametrised functions:

Layers: input $\rightarrow$ hidden $\rightarrow$ output

Each neuron computes:

$$y = \sigma\left(\sum_i w_i x_i + b\right)$$

σ = activation (ReLU, sigmoid, tanh)



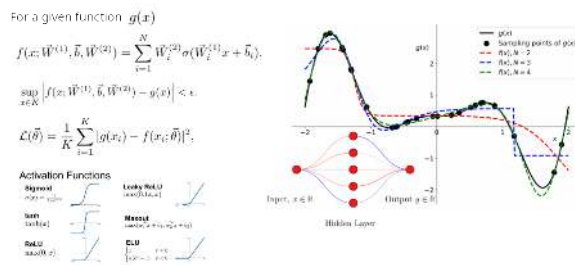
A feed-forward neural network.

Universal Approximation Theorem

Universal Approximation Theorem

A NN with *one* hidden layer can approximate *any* continuous function on a closed, bounded domain — given *enough* neurons and a non-polynomial activation.

(An *existence* result — actually *finding* those weights is another matter.)



More hidden neurons → better approximation of *any* function.

Learning in a Neural Network

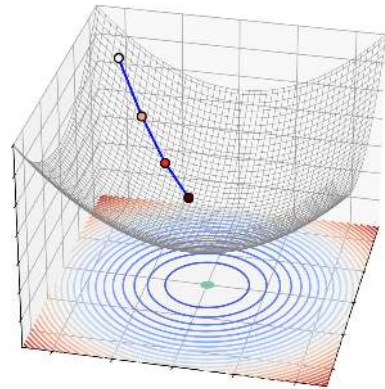
A neural network is just **millions of adjustable knobs** (its weights).

Training, in one idea:

- show it an example → get a prediction
- measure how *wrong* it is — the **loss**
- nudge every knob a little to shrink the loss
- repeat over millions of examples

That downhill nudging is **gradient descent** — rolling toward the bottom of an "error landscape". No rules are programmed; the knobs *settle* into them.

→ Lectures 3–4



Training = rolling downhill to lower error.

The Deep-Learning Boom after 2012 (1/2)

Neural networks have been known since the 1950s (perceptron) and trainable since the 1980s (backpropagation).

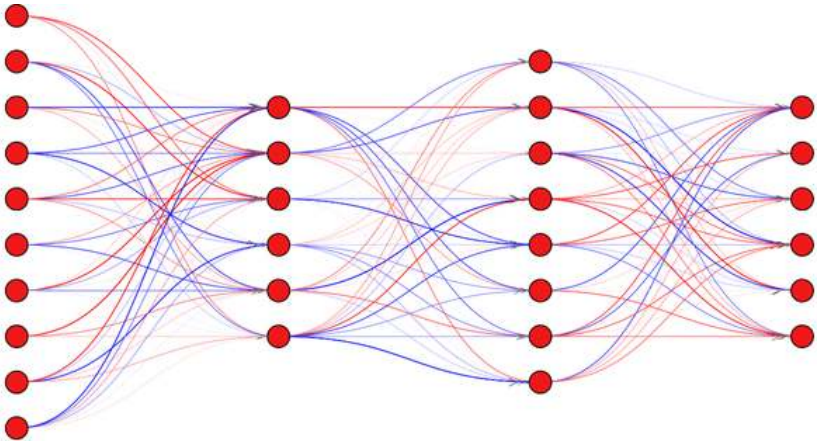
Why the boom only after $\sim$2012?

Enabling factors:

- **Data** — Internet, sensors, digitised archives
- **Compute** — GPUs: built for games, perfect for matrices
- **Software** — Python, automatic differentiation (PyTorch, JAX) computes every gradient
- **Open science** — arXiv, GitHub, shared benchmarks (ImageNet, PDB)

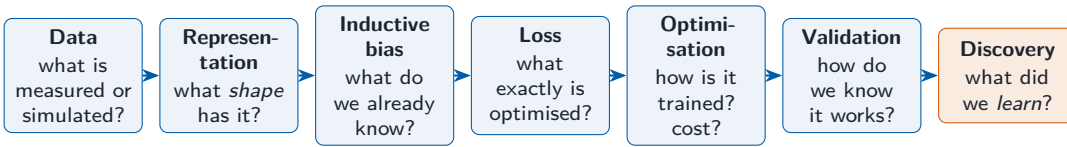
Ch. 1 of Heightman & Płodzień, Springer (to appear, 2027), arXiv:2508.10666

The Deep-Learning Boom after 2012 (2/2)



The deep-learning training loop — the subject of Lectures 2–4.

The Discovery Recipe — Our Lens for the Whole Course



- the same seven questions apply to a student project *and* to AlphaFold
- most failures in published ML-for-science papers hide in **Data** and **Validation**
- **Inductive bias** is where *your* domain knowledge enters: symmetries, conservation laws, chemistry, geography…
- a good prediction is not yet a **discovery**

Your Data Has a Shape — the Architecture Follows (1/2)

Data shape	Examples across disciplines	Architecture	Lecture
table of numbers	lab measurements, patient records, survey data	MLP	2–4
image / grid	microscopy, satellite & radar maps, telescope images	CNN	5
sequence	DNA, proteins, text, time series (climate, finance)	RNN, Transformer	7
graph	molecules, crystals, river & road networks, proteins	GNN	5

Your Data Has a Shape — the Architecture Follows (2/2)

Data shape	Examples across disciplines	Architecture	Lecture
field on a mesh	fluid flow, weather, PDE solutions	CNN, GNN, neural operators	10
no labels at all	large unannotated archives	autoencoders, clustering	6
decisions	control of an experiment, search for algorithms	RL	9

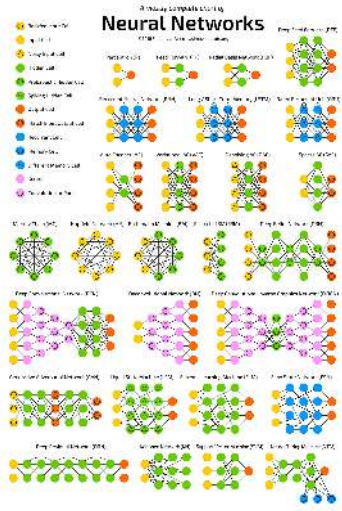
Back to your answer at the start: where does your data sit in this table?

Zoo of Architectures (1/2)

- **FNN / MLP** — tabular data
- **CNN** — images & spatial data (ResNet, U-Net)
- **RNN / LSTM / GRU** — sequences, time series
- **Transformers** — language, any sequence (GPT, BERT, ViT, LLMs)
- **GNNs** — graphs & molecules
- **Autoencoders / VAEs** — compression
- **Generative models** — *generation*: GANs, diffusion, normalising flows

Don't memorise the zoo — learn the *principles* that generate it.

Zoo of Architectures (2/2)



5

A Tour of AI-Driven Discoveries



JAGIELLONIAN UNIVERSITY
IN KRAKOW

The Map: One Course, Many Fields (1/2)

Field	System	What it did	Method, Lecture
Biology	AlphaFold 2/3, AlphaGenome	protein structure from sequence; effects of DNA variants	attention, geometry → 7
Chemistry & materials	GNoME, MatterGen	millions of candidate crystals; materials on demand	GNN, diffusion → 8
Mathematics	AlphaTensor, FunSearch, AlphaEvolve	new matrix-multiplication algorithms & constructions	RL, LLM + search → 9, 15
Physics	tokamak control; fluid singularities	plasma shaping; new unstable blow-up solutions	RL, PINNs → 9, 10

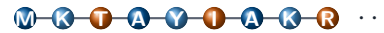
The Map: One Course, Many Fields (2/2)

Field	System	What it did	Method, Lecture
Earth sciences	GraphCast, GenCast	10-day weather forecasts in a minute	GNN, diffusion → 8, 10
Astronomy	DINGO; Kepler CNN	gravitational-wave inference; new exoplanets	SBI, CNN → 5
Everything	LLMs (Claude, ChatGPT), AlphaCode	reading, coding, reasoning assistants	Transformers → 14, 15

Biology: AlphaFold & Protein Folding (1/2)

- **A 50-year grand challenge:** a protein's 1-D amino-acid sequence folds into a *3-D shape* — and **shape determines function**. Predict shape from sequence?
- **Brutally hard:** *Levinthal's paradox* — $\sim 10^{300}$ possible shapes, yet it folds in *milliseconds*.

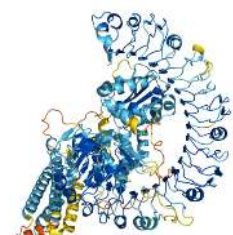
Input — 1-D amino-acid sequence:



(one-letter codes for the 20 amino acids)

AlphaFold ↓

Output — 3-D structure:



Biology: AlphaFold & Protein Folding (2/2)

- **AlphaFold2 (DeepMind, 2020):** a *deep neural network* trained on the **Protein Data Bank** — ∼**170,000** experimental structures gathered over ∼50 years. Nature **596**, 583 (2021)
- **The payoff:** **near-experimental accuracy**, a structure in *minutes*; 200M+ structures *free* → drug & vaccine design; **Nobel 2024**.
- **Next steps:** AlphaFold 3 — complexes with DNA, RNA, drugs Nature **630**, 493 (2024); AlphaGenome — reading the regulatory code of DNA Nature **649**, 1206 (2026)

AlphaFold Through the Discovery Recipe

Data	$\sim$170k experimental structures (PDB) + evolutionary relatives of each sequence (multiple sequence alignments)
Representation	sequence *and* a pair map "residue *i* talks to residue *j*"
Inductive bias	attention between residues; 3-D geometry that respects rotations & translations
Loss	distance between predicted and true atom positions (+ auxiliary terms)
Optimisation	128 TPU cores for $\sim$11 days — far beyond a single lab
Validation	*blind* test: CASP14 competition, structures unknown to anyone
Discovery	a tool, not a law: structures for 200M+ proteins; the *physics* of folding is still not "explained"

We will open each of these boxes in Lecture 7.

Multiplying Two Matrices

A **matrix** is just a table of numbers (e.g. genes × samples, pixels of an image, weights of a neural network)

Rule: each entry of the result = a **row** of the first table times a **column** of the second, summed up.

$$\begin{pmatrix} \mathbf{1} & \mathbf{2} \\ 3 & 4 \end{pmatrix} \begin{pmatrix} \mathbf{5} & 6 \\ \mathbf{7} & 8 \end{pmatrix} = \begin{pmatrix} \mathbf{19} & 22 \\ 43 & 50 \end{pmatrix}$$

$$\mathbf{19} = \mathbf{1} \cdot \mathbf{5} + \mathbf{2} \cdot \mathbf{7}$$

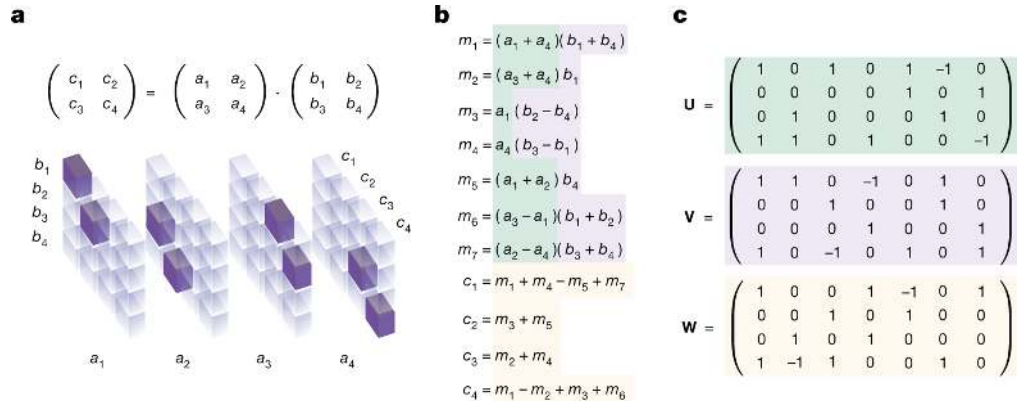
Counting the work:

- each entry: 2 multiplications
- 4 entries ⇒ **8 multiplications**
- an $n \times n$ table: n^3 multiplications ($n = 1000$: a billion)

Why it matters: neural networks and simulations are *mostly* matrix multiplications — fewer ⇒ faster, less energy.

Strassen (1969): the same 2×2 result with only **7** multiplications — a clever trick nobody expected. Can we find more such tricks? → AlphaTensor

Mathematics: AlphaTensor (1/2)

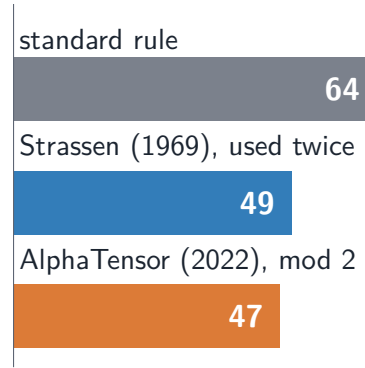


(a) 2×2 matrix multiplication as a 3-D tensor; (b) Strassen's algorithm: 7 multiplications instead of 8; (c) the same algorithm as a tensor decomposition. Fawzi *et al.*, Nature **610**, 47 (2022), CC BY 4.0

Mathematics: AlphaTensor (2/2)

- finding an algorithm = decomposing a tensor into as few pieces as possible — played as a *game* by an AlphaZero-type RL agent
- 4×4 matrices (mod 2): **47** multiplications vs Strassen's **49** — first improvement since 1969
- algorithms tuned to specific hardware: 10–20 % faster on GPUs/TPUs
- **not** a new "fastest" algorithm in general: the asymptotic complexity is unchanged

Multiplications for two 4×4 matrices



Machine-Found Mathematics

- **AlphaTensor (2022)** — RL plays the "decompose the tensor" game.
- **FunSearch (2023)** — an LLM writes programs, an evaluator scores them: new, larger *cap sets* in combinatorics. Nature **625**, 468 (2024)
- **AlphaEvolve (2025)** arXiv:2506.13131 — evolutionary search over code with LLMs: 4×4 complex matrices with 48 multiplications; improved bounds on open geometric problems.
- **AlphaProof** Nature (2025) / **AlphaGeometry** Nature **625**, 476 (2024) — olympiad-level proofs, *verified* by a formal checker (Lean) or a symbolic engine.

The pattern: **generator** (neural network) + **verifier** (exact checker) = results you can *trust*. The output is a *proof* or an *algorithm* — checkable by a human, independent of the network.

Mathematicians: is a machine-found construction "understood"?

Proof Assistants: a Referee for Mathematics (1/2)

A proof is written as a *program*; a small trusted core checks *every* logical step.

Accepted $\Rightarrow$ correct — whoever wrote it, a human or an AI.

Lean: started by L. de Moura at Microsoft Research (2013); community library Mathlib.

AI + Lean: AlphaProof searches for proofs, Lean checks them. With AlphaGeometry 2 at the 2024 Mathematical Olympiad: 4 of 6 problems, 28/42 points — silver-medal level (with multi-day computation).

DeepMind (2024); Nature (2025)

In empirical science the referee is *observation*; in mathematics it is the **proof checker**.

Proof Assistants: a Referee for Mathematics (2/2)

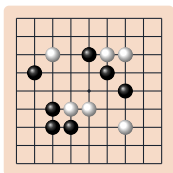
Landmarks of machine-checked mathematics:

- **Four Colour Theorem** — Coq (G. Gonthier, 2005) Notices AMS (2008)
- **Kepler conjecture** on sphere packing — HOL Light & Isabelle (T. Hales *et al.*, Flyspeck, 2014) arXiv:1501.02155
  the same Kepler as in our opening story
- **Liquid Tensor Experiment** — Lean; a challenge by P. Scholze, completed 2022
  blog post
- **PFR conjecture** — Lean, formalised in about three weeks (T. Tao *et al.*, 2023)
  project

Reinforcement Learning (1/2): Games

An agent learns by *trial and error* — solving problems once out of reach.

AlphaGo (2016) / **AlphaZero** (2018) — Go, chess, shogi by *self-play*; AlphaGo's "move 37" — a move almost no human would play.



The game of *Go* ($\sim 10^{170}$ positions).



www.deepmind.com

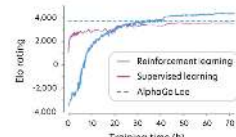


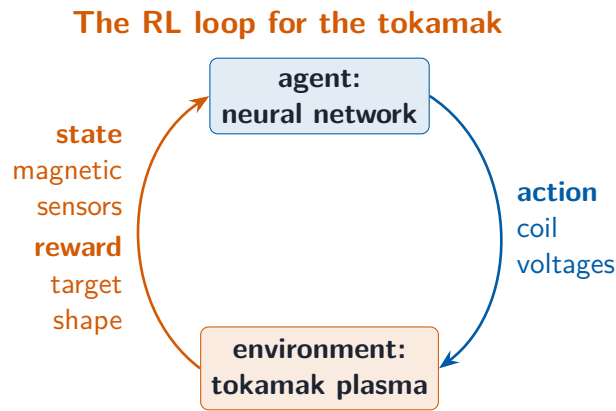
Figure 6.5: Performance comparison between AlphaGo (trained supervised learning) [20], and AlphaGo Zero (pure RL) [309] at the game of Go. Initially, AlphaGo has an advantage thanks to the initial supervised training. However, it learns its capabilities and it is quickly outperformed by AlphaGo Zero. The horizontal dashed line corresponds to the Elo rating of Lee Sedol during his match with AlphaGo being a reference point for the supervised/pure RL performance. Taken from [309].

Mastering the game of Go without human knowledge.
Nature 550 (7676), 354 (2017), doi:10.1038/nature24270.

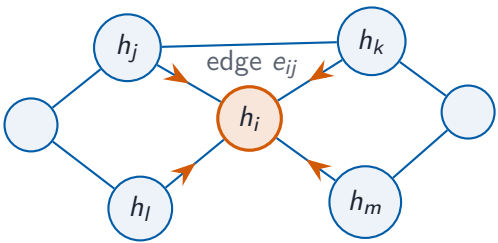
Pure self-play RL (AlphaGo Zero) overtakes supervised learning & the human champion.

Reinforcement Learning (2/2): Science and Engineering

- **Tokamak** — RL controls the magnetic coils that shape a fusion plasma Nature **602**, 414 (2022)
- **Chip design** — RL lays out Google's TPU chips (AlphaChip, Nature **594**, 207 (2021))
- **Robotics** — locomotion & manipulation



Graph Neural Networks in One Picture



each node collects *messages* from its neighbours and updates its state; repeat a few times

Nodes = the objects

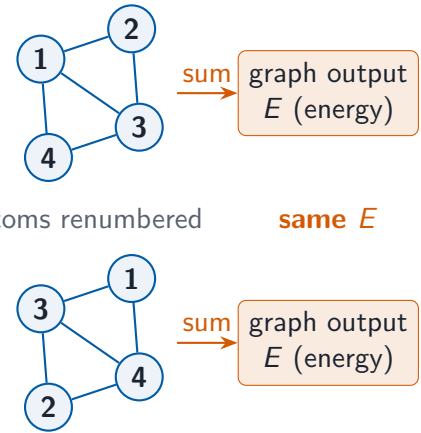
each carries a vector of features h_i: **atoms** (element, charge) · **grid points on Earth** (temperature, wind, pressure) · cities, proteins, particles…

Edges = who interacts with whom

carry relations e_{ij}: **bonds & distances** between atoms · **neighbouring points** on the globe

Graph Neural Networks: Output and Built-in Bias

- **Output:** a property of a node (local forecast) or of the whole graph (energy, stability of a crystal).
- **Built-in bias:** only neighbours talk — locality, and no dependence on how we number the atoms.



Materials & Earth Sciences

GNoME — materials discovery (2023)

Graph neural networks predict the stability of crystals: **2.2 million** new candidate structures, ∼380 000 predicted stable; 736 already made independently by experimental groups. Nature **624**, 80 (2023)

But: chemists questioned how many are genuinely *new* and *useful* — a lesson in validation.

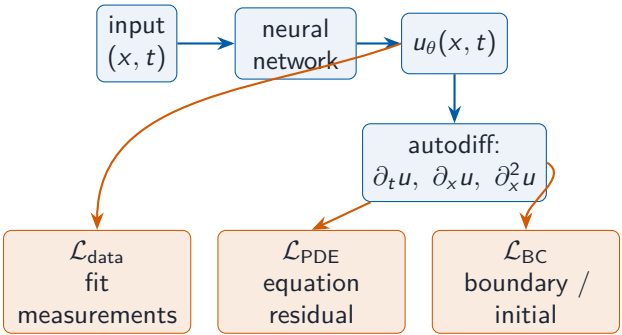
GraphCast — weather forecasting (2023)

A graph neural network trained on 39 years of reanalysis data (ERA5): a 10-day global forecast in **under a minute** on one chip; better than the leading physics-based system on ∼90 % of verification targets. Science **382**, 1416 (2023)

GenCast (2025): *probabilistic* forecasts with diffusion models.

Both learn from data produced by **decades of physics-based simulation** — AI builds *on* traditional science, not instead of it.

Physics: Networks that Obey the Laws of Nature (1/2)



Physics-informed neural networks (PINNs): the equation itself becomes part of the loss → less data needed, physically consistent solutions.

$$\mathcal{L} \;=\; \mathcal{L}_{\text{data}} \;+\; \lambda_1 \underbrace{\left\| \partial_t u + \mathcal{N}[u] \right\|^2}_{\text{physics}} \;+\; \lambda_2\, \mathcal{L}_{\text{BC}}$$

Constraints enter the loss like *penalty / Lagrange-multiplier* terms: conservation laws, symmetries, positivity...

Physics: Networks that Obey the Laws of Nature (2/2)

Singularities in fluid equations (2025)

Do smooth flows blow up in finite time? (Behind the Navier–Stokes Millennium Prize problem.)

PINNs + high-precision optimisation found *new families of unstable self-similar blow-up solutions* for Euler/Boussinesq-type equations — to near machine precision.

Y. Wang *et al.*, arXiv:2509.14185 (2025)

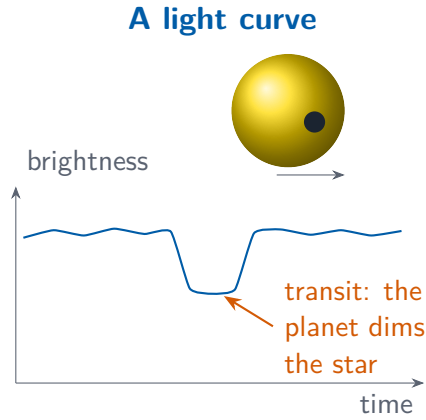
A **self-similar** blow-up keeps its shape while it shrinks and grows towards the blow-up time T:

$$u(x,t) = (T-t)^{-\alpha}\, U\!\left(\frac{x}{(T-t)^{\beta}}\right)$$

- the network learns the profile U; the equation becomes the loss, as for a PINN
- *unstable*: it appears only for finely tuned initial data, so simulating forward in time does not find it

Astronomy: When Only Machines Can Look at Everything

- **Exoplanets:** a CNN on light curves from NASA's *Kepler* space telescope (a star's brightness over time — a planet passing in front dims it) found Kepler-90i — the 8th planet of its system. AJ **155**, 94 (2018)
- **Gravitational waves:** neural posterior estimation infers black-hole parameters in *seconds* instead of hours–days. PRL **127**, 241103 (2021)
- **Sky surveys:** the Rubin Observatory sends ∼10 million alerts per night — only machines can triage them.



Large Language Models in Science

- **Transformers** (2017) → GPT, Claude, Gemini, . . . — trained to predict the next token on most of the written web
- **AlphaCode (2022)** — competitive programming at the level of a median human competitor Science **378**, 1092 (2022)
- 2025: **gold-medal level** at the International Mathematical Olympiad
- daily research use: literature, code, data analysis, writing

Open questions

- do they *understand* or interpolate?
- hallucinations: fluent $\neq$ correct
- can they make *genuinely new* discoveries?
- who verifies the output?

Limits of Deep Learning (1/2)

Limits to keep in mind

- **Correlation, not causation** — a network learns whatever predicts the label, including artefacts
- **Extrapolation is hard** — outside the training distribution all bets are off
- **Data leakage** — found in 294 papers across 17 fields Kapoor & Narayanan, Patterns (2023)
- **Prediction $\neq$ understanding** — a black box does not give you a law of nature

Limits of Deep Learning (2/2): Keeping It Scientific

What makes it scientific

- blind, held-out validation (like CASP)
- an independent *verifier* (proofs, experiments, simulations)
- uncertainty estimates
- open code & data — reproducibility

The *scientist* remains in the loop: choosing data, building in knowledge, judging results.

6
Wrap-up

JAGIELLONIAN UNIVERSITY
IN KRAKOW

Summary

- **AI ⊃ ML ⊃ DL** — deep learning learns the *model itself* from data
- **Three paradigms:** supervised, unsupervised, reinforcement learning
- **Data is the resource** — and its *shape* decides the architecture
- **Real discoveries** in biology, mathematics, materials, physics, Earth sciences, astronomy
- **The discovery recipe:** data → representation → inductive bias → loss → optimisation → validation → discovery
- **A prediction is not yet a discovery** — validation is the scientist's job

A Few Words of Advice

1. Dive in

Pick a topic you *are curious about* — and go deep.

2. Everything connects

Everything you learn can become useful — the unexpected connections are the norm.

3. Embrace not-knowing

Enjoy learning. Accept *not understanding*; be at peace with *not knowing — yet*.

4. Research is joy

Revealing the *unknown*, solving the *unsolved*, facing the *hard* — that is what makes research so exciting.

Pick a small problem. Reproduce a paper. Then improve it. You don't need permission — **curiosity is enough**.

Thank You

Questions, project ideas, feedback: marcin.plodzien@uj.edu.pl

Website · GitHub · Google Scholar



